

Descending Pathways to the Spinal Cord, IV: Some Factors Related to the Amount of Cortex Devoted to the Corticospinal Tract

R.J. NUDO AND R.B. MASTERTON

Department of Psychology, Florida State University, Tallahassee, Florida 32306-1051

ABSTRACT

In the companion paper to this one (Nudo and Masterton: *J. Comp. Neurol.* 296:559-583, '90), we have presented data indicating that in each of 22 mammals, there are either 2 or 3 separate regions of neocortex contributing corticospinal fibers. In this paper, we describe the variation in the absolute size of these cortical regions, the total amount of neocortex contributing corticospinal fibers (CST cortex), and the total amount of neocortex (total cortex) in each of the animals. We then use strict statistical tests to examine the relationships between these measures and several other quantitative measures or descriptions of the animals' size, ancestral heritage, motor prowess, and ecological adaptation.

The results show that the absolute amount of CST cortex is more closely related to the total amount of neocortex than to any other quantitative measure available. The further variation—that is, the variation in the amount of CST cortex relative to total neocortex—appears to have been random over the inferred ancestral lineages of most animals in the sample, but seems to have been almost absent along the anthropoid lineage. Because this constancy in the relative amount of CST cortex over a very long period of anthropoid ancestry is apparently unusual if not unique among mammals, it may contain a clue to the special role of the corticospinal tract among primates.

Finally, the distribution of the CST among the 3 cortical regions in primates was found to be more closely related to their particular mode of ecological adaptation than to their particular combination of digital dexterity and hand-eye coordination.

Key words: motor system, evolution, cortex

In the companion paper to this one, we have described the commonalities and qualitative variations of the cerebral neocortical regions originating fibers of the corticospinal tract (Nudo and Masterton, '90). Here we report the absolute and relative areas of these cortical regions in the 22 species of mammals in the sample. We then illustrate the outcome of strict statistical tests of several commonplace notions regarding the morphological, taxonomic or evolutionary, functional, and ecological characteristics that help to account for the variation in the amount of cerebral cortex devoted to the CST.

MATERIALS AND METHODS

The procedures for retrogradely bulk-labeling the cells originating the corticospinal fibers have been described in detail elsewhere (Nudo and Masterton, '89). Briefly, with the animal under deep Chloropent anesthesia we applied flakes of raw HRP to the freshly cut fibers of the spinal cord after it had been hemisected at the C1-C2 junction (see Hicks and D'Amato, '77). After a 3 day survival, the animal

was deeply anesthetized once more and after perfusion and sectioning, we processed the tissue with the TMB procedure and then counterstained some sets of sections with either cresyl violet or neutral red.

In the previous paper we described two separate regions, and in 12 of the 22 species, three separate regions of neocortex originating CST fibers (Nudo and Masterton, '90). In the present paper we focus on the size of these cortical regions and their relationships with other characteristics of the animals.

Area measurements

In order to measure the areas of the cortical regions originating CST fibers, we used a computerized planimeter on optical projections of systematic sets of coronal or sagittal sections that had been HRP-processed and cresyl-counterstained. For each section we measured the length of

Accepted December 21, 1989.

R.J. Nudo's present address is Department of Neurobiology and Anatomy, University of Texas Health Science Center, Houston, TX 77225.

the cortical surface above the patches of HRP-labeled cells in the hemisphere containing the maximum number of cells (i.e., in the contralateral hemisphere in all but the hedgehogs and mole).

In addition, we measured the dorsal-ventral length of the entire neocortical surface from its border with the cingulate cortex to the depths of the rhinal fissure (in coronally sectioned brains) or from its rostral to its caudal limits (in sagittally sectioned brains). By multiplying the distance between the sections by the sum of each set of planimetric measurements, we obtained estimates of the areas of the entire neocortex ("total cortex") and for each region originating CST fibers. By summing these latter measurements, we also derived the total area of cortex originating CST fibers ("CST cortex").

We measured every 10th 50 μ m section (i.e., sections 500 μ m apart) for the large brains (i.e., primates and carnivores), every fifth section (i.e., 250 μ m apart) for smaller brains (e.g., most rodents, *Didelphis* opossum), and two sets of every fifth section (i.e., averaging 125 μ m apart) for the very smallest brains (e.g., Insectivores, *Monodelphis* opossum). The measurement error inherent in this technique is less than 3% for regions A and B and for total neocortex, less than 10% for regions C and C', and less than 5% for the area of cortex devoted to the CST. We return to this point below.

Parameters examined statistically

Allometric relationships. We began our statistical analysis by examining allometric or scaling factors to see if the area of the cortex giving rise to corticospinal fibers may have a direct relationship with either body size, brain size, or total neocortical area.

Taxonomic and phylogenetic relationships. The second set of parameters examined consisted of those describing the animals' heritage, in particular their degree of kinship with mankind and then, with raccoon and tree squirrel because of their superficial similarities with primates either in motor dexterity or in good vision, respectively. Our intention was to determine how much of the variation in CST cortex area could be accounted for by the differing ancestral lineages without reference to the animal's motor or sensory skills.

The technique for assigning numbers to the various species designating their relative kinship (or propinquity or relative recency of last common ancestor, etc.) has been described in detail previously (Nudo and Masterton, '89). Briefly, the "relative kinship" scaling relies on cladograms by others that provide the sequence of major branch points in the phylogenetic lineages of the animals in the sample (e.g., Wiley, '81). For the present purposes, we found cladograms tracing the lineages of humans, raccoons, and tree squirrels in sufficient detail to yield at least five-step scales of their ancestral lineages (McKenna, '75; Eisenberg, '81).

Figure 1 shows the cladogram and phylogenetic relationships among the animals mentioned in this report. For kinship ranking, it shows that the two marsupials have been arbitrarily assigned the lowest kinship rating of "2" because among the animals in the present sample they have the most remote common ancestry with either mankind, raccoon, or squirrel. (The kinship ranking of "1" has been reserved for reptiles, which were included in previous reports but do not appear in this one, e.g., Nudo and Masterton, '89.) At the other end of the kinship scale for mankind, the three Old World monkeys have the most recent common ancestry and

are ranked "6," the highest number. The 13 non-primate placentals including tree shrew, two prosimian primates, and two New World monkeys are ranked "3," "4," and "5" respectively, because of their successive points of last common ancestry along the lineage to humankind (see Fig. 1).

For the scaling of kinship with raccoon (see circles in Fig. 1), "3" is assigned to the 18 non-Carnivore placentals, "4" to the cat, and "5" to the raccoon itself. For tree squirrel, "3" is assigned to the nonrodent placentals, "4" to rat and vole, "5" to ground squirrel, and "6" to the tree squirrel itself (see horizontal ovals in Fig. 1). Finally, for European hedgehog, "3" is assigned to the non-Insectivora placentals, "4" to mole, "5" to least shrew, "6" to *Hemiechinus*, "7" to *Erinaceus albiventris*, and "8" to *E. europus* itself (see vertical ovals in Fig. 1).

For a second type of phylogenetic analysis best explained as the results unfold, the characteristics of the sequence of ancestral species at the successive branch points along the same lineages were estimated by using the unweighted average of the successive subsets of their respective progeny (see Nudo and Masterton, '89). In this procedure the trait average is cumulated beginning with the right-most (or most recent) set of progeny and working backward to the least recent.

Motor skill. After allometric and heritage analyses, we examined two measures of motor prowess. Because of good physiological (Lawrence and Kuypers, '68) and comparative reasons (Heffner and Masterton, '75, '83) for suspecting "dexterity" or "independent movement of the digits" as a primary function of the primate CST, the first of these was "digital dexterity."

The present six-point scaling of this parameter follows the ordinal scale of Napier and Napier ('67) as extended to non-primates by Heffner and Masterton ('75, '83). In this scaling, the mammals are rated from 1 to 7 mostly on the anatomy of their forepaw (convergent digits, prehensile digits, opposable thumb, etc.). This scale ranks the "precision grip" of humans as "7" and the primitive pentadactyl of mammals as "3." Animals with hooves and fused digits such as moles (or horses and cattle, etc.) are ranked "1" and those with paws and partly fused digits such as hedgehogs, rabbits, cats, and marmosets are ranked "2." For the present purpose, this scale has been applied with only one modification—a scaling up of raccoon to a dexterity level equaling monkeys (see Heffner and Masterton, '83).

The second scale of motor prowess analyzed explicitly was "hand-eye coordination," borrowed from our previous analysis of the tectospinal tract (Nudo and Masterton, '89). For this parameter we derived an unrefined three-point scale ranging from 0 to 2 where "0" denotes animals that can not, or normally do not, view their forepaws, "2" denotes animals with good hand-eye coordination (monkeys, cats, raccoons), and "1" denotes all intermediate cases (Nudo and Masterton, '90).

Therefore, the two dimensions, digital dexterity and hand-eye coordination, both admit of unambiguous (though entirely arbitrary) scaling among the mammals of the present sample. Although we anticipate including other measures too (e.g., of speed and strength), at present these measures are not available for most of the animals.

Ecological relationships. The final set of analyses was focused on the general lifestyle of the animals: food preference (carnivorous, herbivorous, omnivorous, insectivorous), preferred nesting site (burrow, surface, tree), usual foraging location (fossorial, terrestrial, arboreal), and their

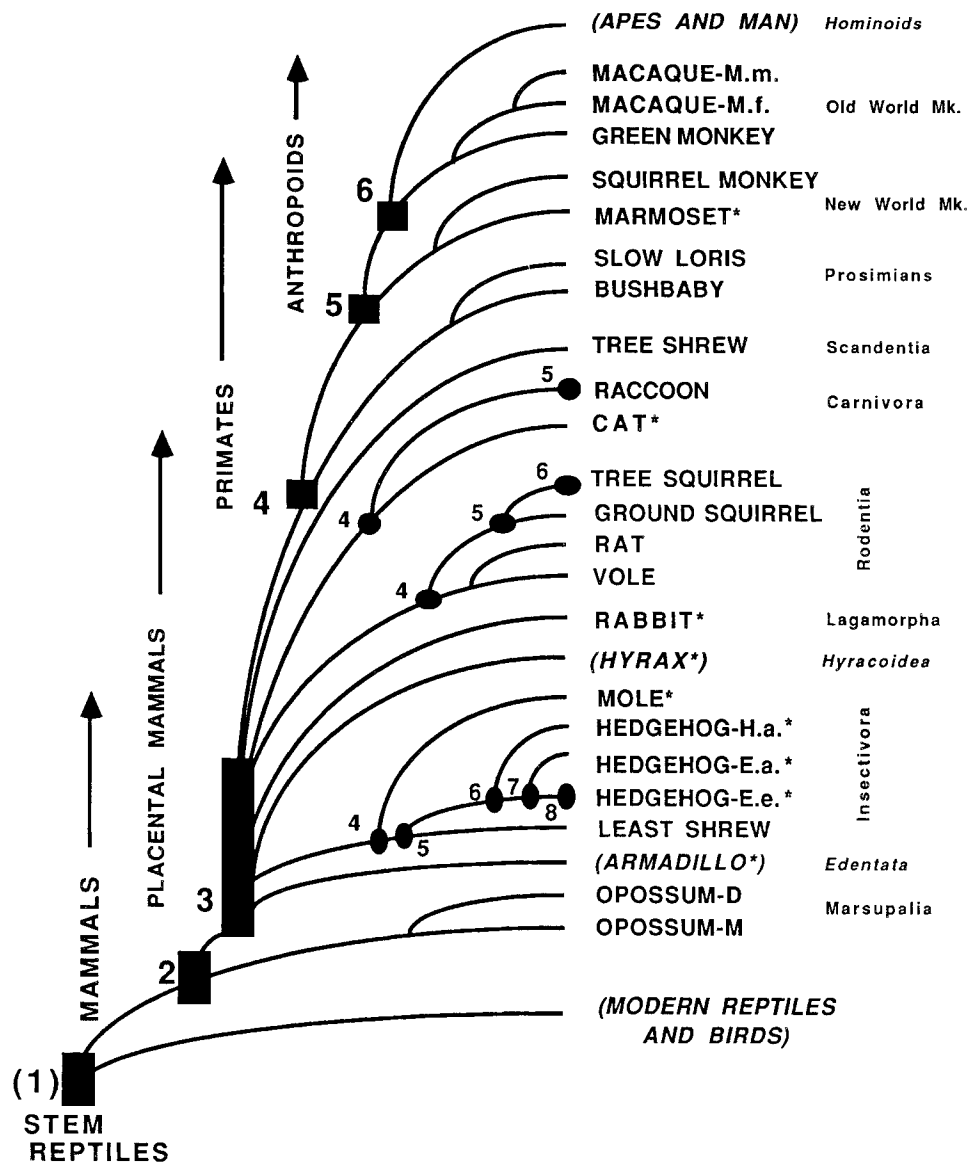


Fig. 1. Cladogram showing phylogenetic relationships of species mentioned in text. Names of animals in parentheses and italics are not included in formal sample. Labels at far right indicate orders for nonprimates, suborders or families for primates. Asterisks indicate animals with fused or partly fused digits. Numbered symbols label suc-

cessive (extinct) ancestors whose characteristics can be inferred from characteristics of (extant) progeny in sample. Vertical Rectangles: ancestors common to each lineage of placental mammals in sample; Squares: anthropoid ancestry; Circles: raccoon ancestry; Horizontal Ovals: tree squirrel ancestry; Vertical Ovals: hedgehog ancestry.

trophic level (more likely prey or predator). Because we could find no basis for a numerical scaling of these characteristics, we relied on "analysis-of-variance" techniques for their analysis (see below).

Statistical techniques

We have used two statistical techniques, regression and analysis-of-variance, to analyze possible relationships between the several quantitative measurements. Although these techniques are used here in an entirely conventional fashion, it should be noted that neither is the strongest or most refined variant of the technique that might have been used. Furthermore, the arbitrary specification and scaling of some characteristics, such as motor prowess and cladistic

kinship, with simple ordinal scales are, in each case, the *least* refined estimates we could choose.

We chose these crude scalings and unrefined statistical techniques on purpose. Our object was to be sure that the conclusions derived here did *not* depend on subtle or arguable distinctions nor on unfamiliar statistical techniques whose applicability would have to be taken on faith even if mathematically correct. The outcome of these choices means that each of the analyses that yields a statistically reliable relationship here is very likely to remain statistically reliable when other more refined statistical techniques are used or more refined scalings are derived. (In statistical terms, we have risked Type II errors in order to minimize Type I errors.)

Therefore, it is only the strongest or statistically most obvious relationships that are discussed here. More subtle relationships between the several dimensions, if they do exist, would pass undetected by the methods chosen here and be omitted entirely from the conclusions. Therefore, the *nonreliability* of a relationship should not be construed as proof that it does not exist, nor that it would not become reliable if a larger or better sample of mammals became available.

RESULTS

Figure 2A lists the 22 mammals in the sample with measures of the area of each cortical region, the amount of cortex originating CST fibers (i.e., the sum of the areas of

region A, B, and C or C' where either is present), and the total neocortical area. Within the list, the animals are arranged in descending order of the percentage of cortical area devoted to the origin of corticospinal fibers—that is, the last column in Figure 2A. Figure 2B–D shows the distribution of this percentage among the 22 species, 14 families, and seven orders represented in the sample. It can be seen from these distributions that for this particular trait, primates ("P" in Fig. 2B,C) are quite ordinary mammals.

Allometric relationships

From the table in Figure 2A, allometric analyses were performed by calculating least-squares, regression equations, and correlation coefficients first among the six col-

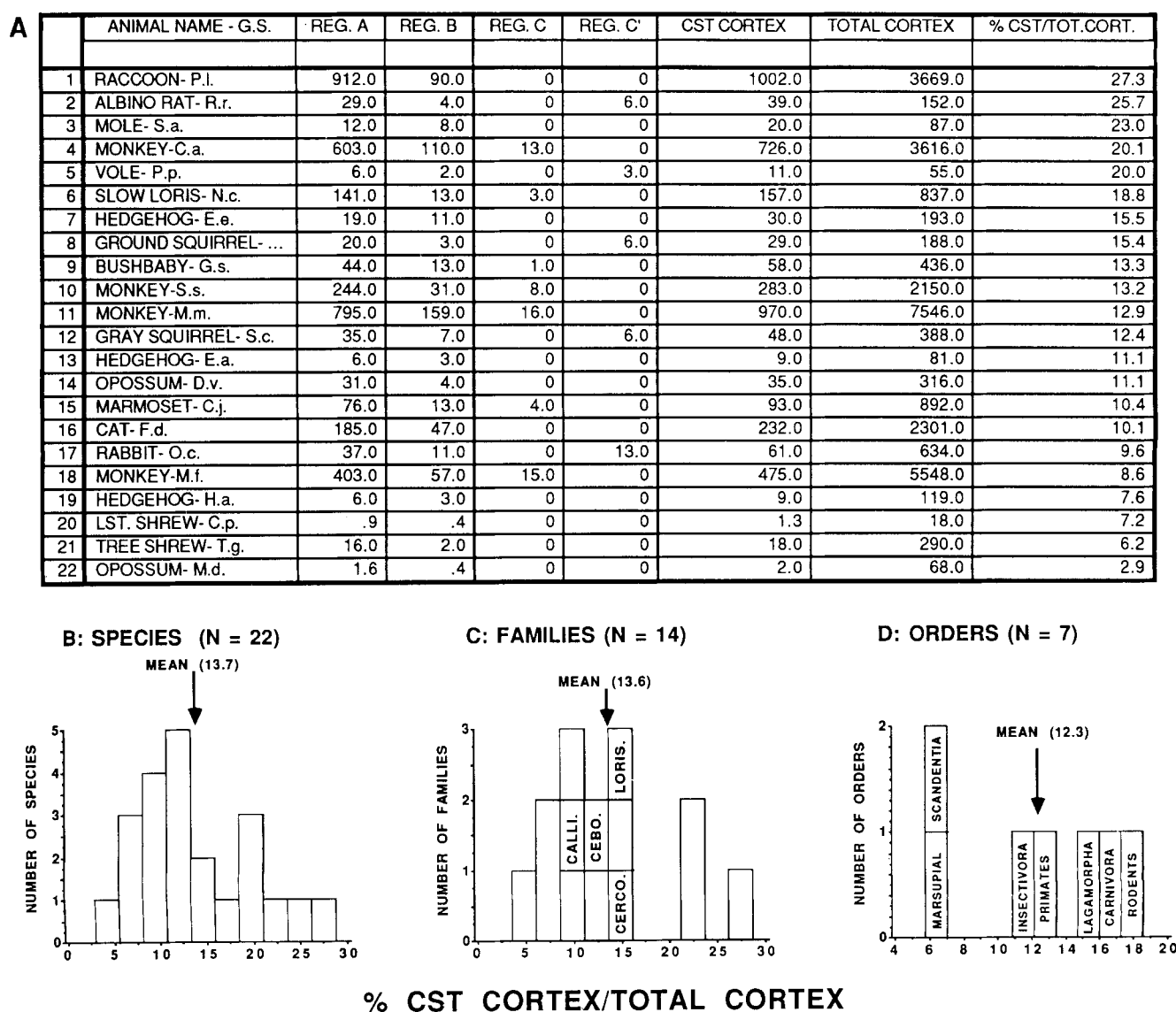


Fig. 2. **A:** Areas (in sq. mm.) of cortical regions originating corticospinal fibers, their sum ("CST cortex"), and area of total neocortex ("total cortex") for each of 22 mammals. Animals are listed in descending order of relative amount of CST cortex (last column of table). **B-D:** Distribution of relative amount of CST cortex among the 22 species (B),

14 families (C), and seven orders (D) represented in sample. Symbols within B and C are the seven species and four families of primates represented. Note that primates cluster around mammalian average in each distribution suggesting that for this particular trait they are not unusual mammals.

umns of data and then between each column vs. brain weight and body weight, and then once again among each of their respective logarithms. Since in each case here and in each analysis that follows, the logarithms of the measurements were more closely related than the raw measurements themselves, only logarithmic relationships will be discussed further.

Graphs of the relationships between the log of the area of CST cortex and log body weight, log brain weight, and log total cortical area are shown along the top row of Figure 3. It can be seen that each of these measures is reliably related to the area of CST cortex, with total neocortex the most closely related (graph C in Fig. 3). By itself, total neocortical area can account for more than 92% of the variation in the cortical area devoted to the CST.

In graphs D and E in Figure 3, the relationships between the area of CST cortex and either body weight or brain weight are regressed after the chief correlate, total cortical area, is mathematically removed as a source of variation. The object of these graphs is to show that although body weight and brain weight are indeed related to the area of CST cortex (graphs A and B in Fig. 3), the predominant relationship is with total cortical area (graph C in Fig. 3) and when this one dimension is removed mathematically, the relationship of CST cortex with either body weight or brain weight falls to chance (graphs D and E in Fig. 3).

Finally, graph F in Figure 3 shows that the close relationship between total neocortical area and the area of cortex devoted to the CST remains close and statistically reliable among the several orders even though the sample size is now reduced to seven.

To summarize the allometric relationships, a large area of cortex devoted to originating corticospinal fibers in the mammals of the sample is *not* directly related to a large body or to a large brain. Instead, it is a characteristic accompanying a large neocortical surface area. Apparently body size and brain size are each secondarily related to the amount of CST cortex because they are also related to total neocortex. We return to the significance of this observation below.

For present purposes, however, because of the close correspondence between CST cortex and total neocortex among each of the animals in the sample including the primates, it is the ratio (or percentage) of CST cortex to total cortex which admits other sources of variation among the animals and warrants further analysis.

Phylogeny of CST cortex

Figure 4 shows the relationships between CST cortex (left), total neocortex (middle), and percentage CST cortex (right) to the relative recency of kinship with mankind

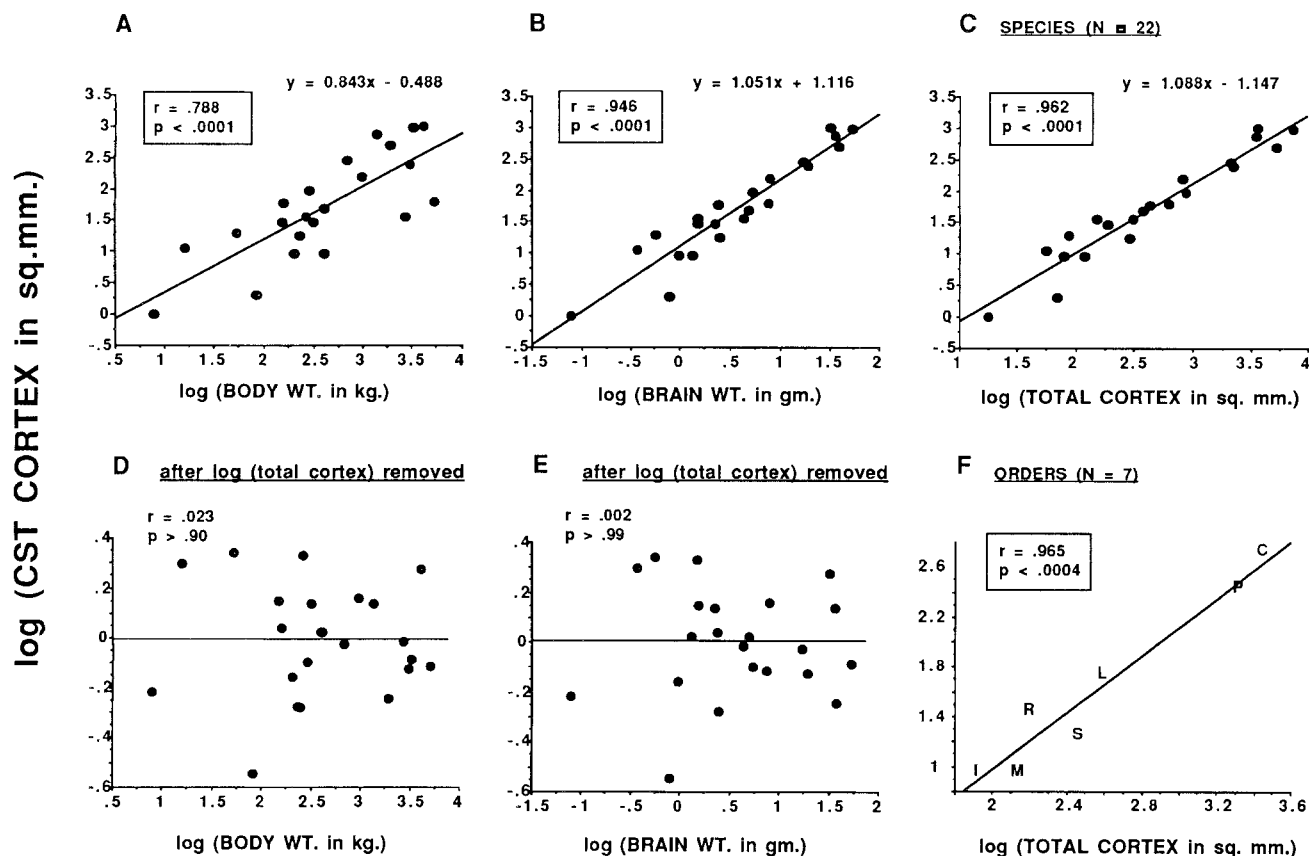


Fig. 3. Allometric relationships of absolute area of CST cortex with body size (A), brain size (B), and total neocortical area (C). D and E show that relationships in A and B collapse entirely after "total neocortical area" has been mathematically removed as a factor. F shows same relationship as C but with animals sampled at the ordinal instead of species level. Note that each parameter is logarithm of a raw measure-

ment and graphs are log-log displays. For these and every other relationship found to be statistically reliable, logarithms of measures were more closely related than the raw measures themselves. In this and following graphs, line is least-squares fit; correlation coefficients (r) and associated probabilities (p) are boxed when statistically reliable, unboxed when not reliable.

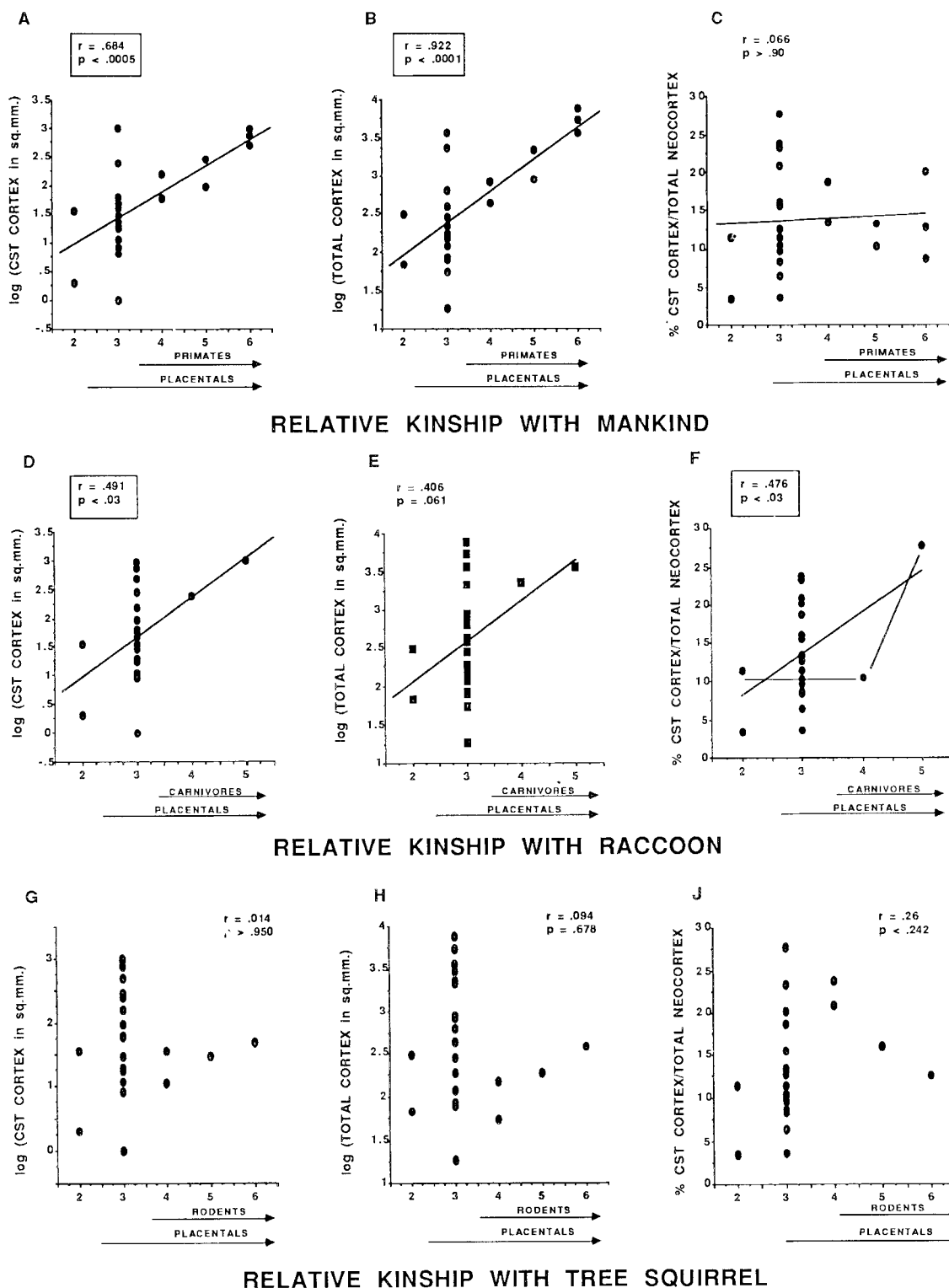


Fig. 4. Relationships of absolute area of CST cortex (left column), total cortex (middle), and percentage CST cortex (right) with relative kinship (or relative recency of last common ancestor) in lineages of mankind (top row), raccoon (middle), and tree squirrel (bottom). Graph in D shows that expansion of CST cortex along mankind's lineage (graph A) was not unique: it expanded along the raccoon lineage as

well. In F, lines show two equally likely alternatives for the timing of the expansion of CST cortex in the raccoon lineage. Note relative constancy in percentage CST cortex along mankind's lineage (i.e., see almost horizontal regression line in C despite large and reliable changes in the amount of CST cortex (A) and total neocortex (B)).

(upper row of graphs), raccoon (middle row), or gray tree squirrel (lower row).

Figure 4A and B shows that the absolute amount of cortex devoted to the CST and the total amount of neocortex increased markedly along mankind's ancestral lineage. However, graph D in Figure 4 shows that the absolute amount of cortex devoted to the CST also increased along raccoon's ancestral lineage. Therefore, Figure 4D shows that the large absolute increase in CST cortex seen in the primates (Fig. 4A) was *not* unique to that lineage—at least the lineage leading to raccoon showed a similar increase.

Although the increases in CST cortex and in total cortex seen in the lineages leading to mankind and to raccoon are similar, they are not identical in detail (compare Fig. 4A,B with C,D). Graphs B, C, E, and F in Figure 4 show how the two lineages probably differed. Along the primate lineage, CST cortex and total neocortex expanded in very close correspondence (graphs A and B are nearly parallel) with the result that the percentage of cortex devoted to the CST remained relatively constant (graph C in Fig. 4).

In the raccoon's lineage, in contrast, total neocortex did not expand in as close a correspondence with CST cortex (graphs D and E) with the result that the percentage of CST cortex shows a late and abrupt increase within the Carnivora segment of the lineage (graph F in Fig. 4).

In contrast to the primate and Carnivora patterns, the bottom row of graphs in Figure 4 shows that in the tree squirrel's lineage, there was no reliable long-range change in either the absolute or relative amount of CST cortex.

Figure 5 shows the changes in the relative amount of CST cortex that probably took place along four lineages. These graphs differ somewhat from the "relative kinship" (or "propinquity") graphs in Figure 4. Instead, they show the percentage of CST cortex averaged over the entire set of progeny of each hypothetical ancestor (or branch point) along each lineage (see Nudo and Masterton, *in press*). In a statistical sense, these averages are the unbiased estimate of the trait for each ancestor. Therefore, the graphs in Figure 5 indicate the most likely changes in the relative amount of CST cortex taking place along each lineage.

To begin with, because the lineages did not diverge until after their common ancestry became Eutheria or placentals, the estimated values are the same in each graph over the first three points (see open symbols in Fig. 5). From that point onward, however, the lineages diverged (see Fig. 1) and Figure 5 shows that the histories of CST cortex began to diverge too (see filled symbols in Fig. 5).

The filled symbols in each graph of Figure 5 show that the further histories of CST cortex probably varied markedly. For example, in the most recent part of the raccoon's lineage, the percentage of CST cortex rose abruptly; in the squirrel's lineage, however, the percentage first rose, then fell, while in the hedgehog's lineage, it first fell and then rose.

In sharp contrast to these unpredictable variations in the lineages of raccoon, tree squirrel, and hedgehog, the percentage of CST cortex in humankind's lineage remained almost constant (Fig. 5A). In view of the historical fluctuations in CST cortex among the other 19 lineages represented in the present sample, this near constancy in relative amount of CST cortex along the early mammalian and early primate segments of mankind's lineage appears singular and may be unique. It is discussed below.

Given the constancy of the relative amount of CST cortex over mankind's lineage, the question turns to possible

historical changes in the distribution *within* CST cortex, that is, among the constituent regions. Figure 6 shows the relationships between kinship with mankind and both the absolute and relative areas of each of the three separate regions of cortex originating CST fibers (Regions A, B, and C). It can be seen that although the *absolute* size of the regions expanded throughout the lineage (top three graphs in Fig. 6), only Region C also shows a statistically reliable increase in *relative* size (bottom three graphs in Fig. 6).

When the relationships shown in Figure 6 are combined with those shown in Figure 5, it can be concluded that the *absolute* amount of cortex devoted to the CST probably increased in several mammalian lineages including human's and raccoon's. However, an increase in the *relative* amount of cortex devoted to the CST was probably *not* a feature of human evolution though it seems to have occurred at a recent stage of raccoon evolution and probably fluctuated during the squirrel's and the hedgehog's lineage.

Motor prowess

Digital dexterity. Figure 7 shows the relationship between digital dexterity and each of four morphological descriptions of the CST: first, absolute area of CST and total cortex as measured here and shown in Figure 2A and then, two other parameters taken from a previous report and based on orthograde tract-tracing studies of others: the ventral-most lamina of the spinal cord gray containing CST terminals and the distance the CST extends down the spinal cord (Heffner and Masterton, '75, '83). In Figure 7, it can be seen that each of these parameters alone is reliably related to digital dexterity even though only small subsets of the present sample are represented in the two lower graphs—15 animals in graph D and nine in graph C. From these graphs, it can be concluded that several characteristics of the CST including its size and pattern of terminations are reliably related to digital dexterity.

Figure 8 shows the relationship of digital dexterity to the relative sizes of the constituent regions of CST cortex. It can be seen that Regions A and C are reliably and directly related to dexterity (see graphs A and C in Fig. 8) while Region B is inversely related (graph B), and Region C' (graph C') is not reliably related at all. Neither of these relationships is particularly strong, each alone accounting for less than 35% of the variation in dexterity. Furthermore, taken together, the variation in relative size of Regions A, B, and C (i.e., the variation in the distribution of the CST among the three regions) accounts for 64% of the variation in dexterity.

Hand-eye coordination. The relationship of CST morphology to a second measure of motor prowess, hand-eye coordination, is shown in Figure 9. Despite its unrefined scaling here, hand-eye coordination is reliably related to the log of CST cortex, total cortex, absolute area of Region A, and relative area of Region C (see graphs A–D in Fig. 9). However, the two closest relationships of any we could find are those shown in graphs E and F in Figure 9: the log of Non-CST cortex (graph E) and the distance the CST extends down the spinal cord (graph F). Taken together, these last two factors alone account for over 85% of the variation in hand-eye coordination in the present sample of mammals and further, they entirely obviate both the relationships shown in the upper graphs—that is, when they are mathematically removed, the relationships between hand-eye coordination and the other available morphological measures disappear.

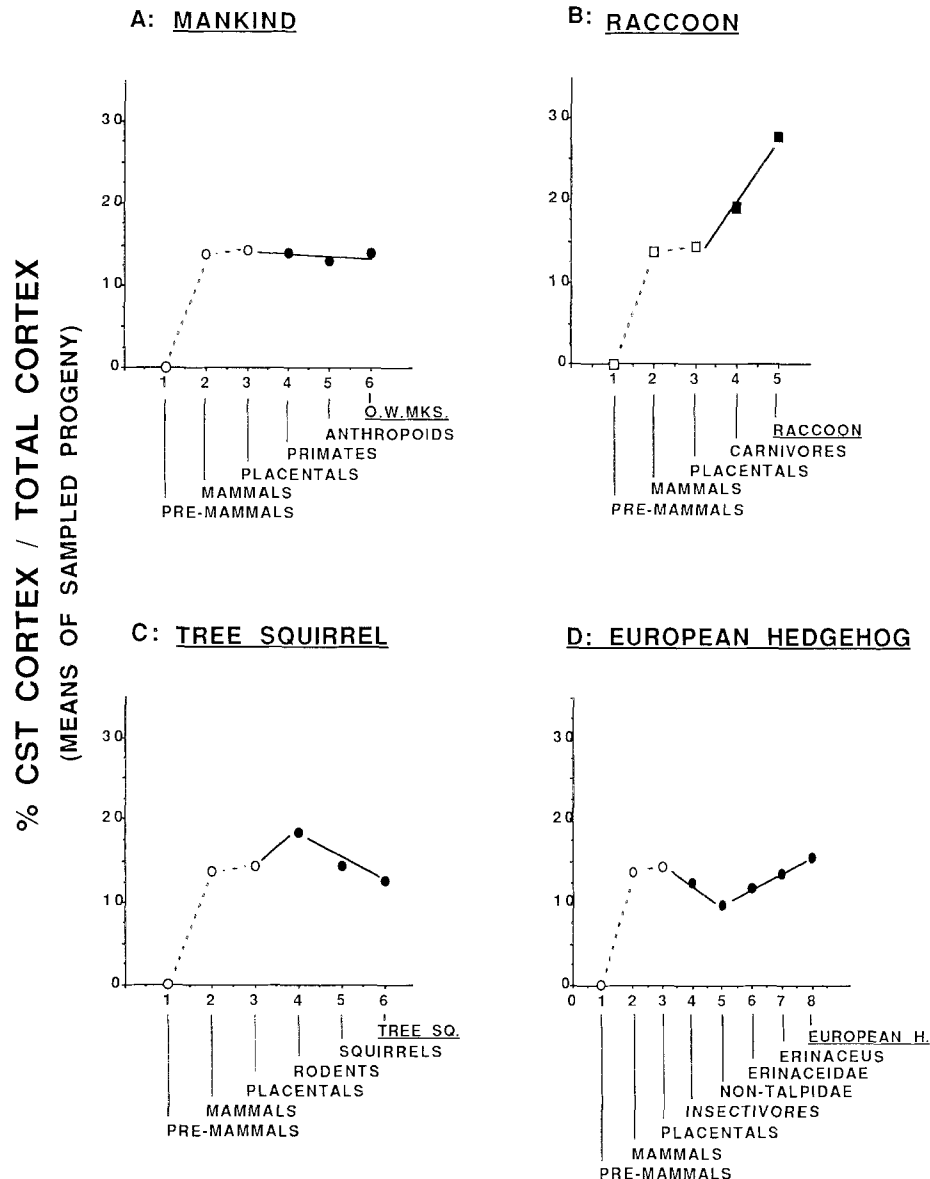


Fig. 5. Current best-estimates of relative amount of CST cortex among some of the successive extinct ancestors along the lineages leading to either mankind, raccoon, tree squirrel, or European hedgehog. Because the first three ancestors are common to each of the lineages (open symbols), the lineages diverge only after the point labeled

"Placentals" (see Fig. 1). Note that this is also the point where the graphs begin to diverge (filled symbols). Note also that the relative amount of CST cortex had a markedly different history along each lineage.

Ecological contributions

So far we have seen that the distribution of CST origins among the cortical regions seems to be related only to digital dexterity and not very closely to that (see correlation coefficients in Fig. 8). Here we turn to the ecological dimensions of the animals' lives—at least those that might place selective pressure on the animals' motor systems. Toward this end, we tested food preference, nesting preference, foraging preference, and trophic level first, against percentage CST cortex relative to total neocortex and then, percentage of Regions A, B, C, and C' each relative to total CST cortex. By this means we hoped to detect at least gross ecological influences on the distribution of CST neurons among the cortical regions.

The principal results are shown in the six graphs in Figure 10. Figure 10A shows that whether or not an animal is an *arboreal forager* is reliably reflected in the percentage of CST cortex found in Region B.

Figure 10B shows that whether or not an animal is *insectivorous* accounts for a large part of the variation seen in the relative area of Region B. The graph shows that although there is overlap in the insectivorous and non-insectivorous subsamples, Region B is relatively larger in enough of the insectivorous species that the probability that the two subsamples are the same is less than 1/10,000.

Figure 10C and D shows that *nesting preference* (fossorial, terrestrial, arboreal) has a reliable effect both on Region C and Region A. The first of these relationships is the

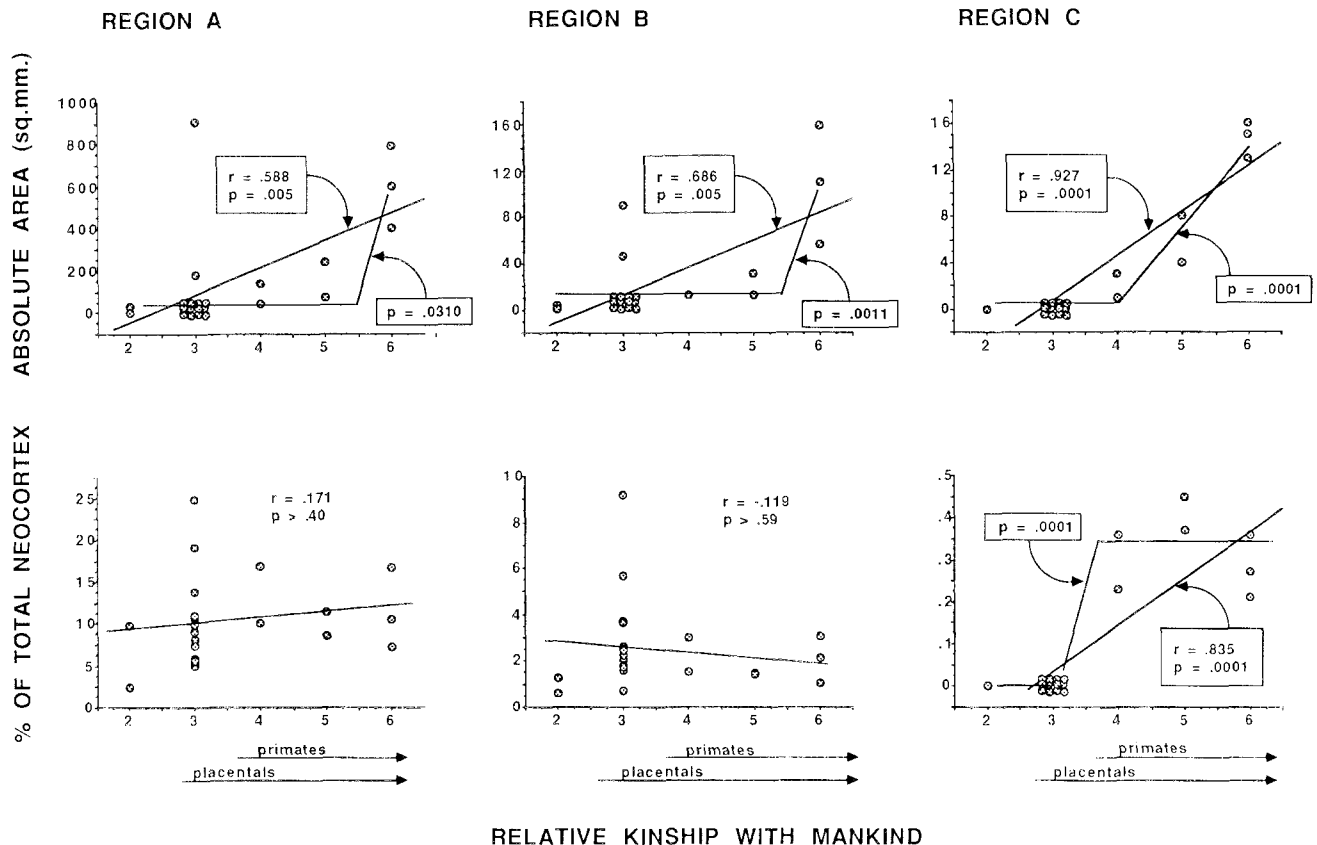


Fig. 6. Absolute (top row) and relative areas (bottom row) of the three major regions of CST cortex along mankind's ancestral lineage. Two lines in same graph show equally likely alternatives for summarizing the histories. Three upper graphs show that the increase in absolute

area is statistically reliable for each of the three cortical regions. However, lower graphs show that only Region C changed in size relative to total neocortex and even then the change may not have been monotonic.

obvious one between having a Region C and being an arboreal nester, since having a Region C is tantamount to being a primate. In Figure 10D, the relationship between nesting preference and Region A depends on burrow-nesting vs. non-burrow (terrestrial or arboreal)-nesting. Here, it is the burrow-nesters (moles, hedgehogs, rodents, etc.) that deviate by having a relatively small Region A. Taken together, Figure 10C and D suggests that both Region A and Region C are influenced by an animal's preferred nesting site—with tree-nesters having a larger Region C than non-tree-nesters and non-burrowers having a larger Region A than burrowers.

Finally, Figure 10E and F shows that an animal's *trophic level*, the likelihood of its being prey or predator, has an effect on both Region B and Region C', with the prey animals having a relatively small Region B and a relatively large Region C'.

Summarizing the relative size of Region A is most closely related to whether its host nests somewhere other than a burrow (Fig. 10D); the relative size of Region B depends on whether the host is an insectivorous predator (Fig. 10B,E); the relative size of Region C depends mostly on an arboreal habit, both in nesting and in foraging (Fig. 10A,C); and finally, the relative size of Region C' seems to be related to a non-predatory trophic level.

Taken together, the four ecological dimensions can account for 44.8% of the variation in Region A, 61.6% in

Region B, 63.3% in Region C, and 57.4% in Region C'. Despite these relatively reliable relationships with the regions taken one at a time, the ecological dimensions can account for only 4.6% of the variation in the relative size of the sum of the regions, that is, in the relative amount of CST cortex. Therefore, the four ecological parameters tested here apparently contribute to the distribution of the CST among the cortical regions without contributing reliably or significantly to the relative amount of CST cortex itself.

DISCUSSION

The results show that the variation in the absolute amount of CST cortex is almost entirely (92%) accounted for by the variation in the absolute amount of cerebral neocortex. The remaining variation, or the relative amount of cerebral cortex devoted to the corticospinal tract, can be accounted for by variations in phylogeny and by parallel variations in motor prowess. In addition, the results show that the distribution of CST origins among the several cortical regions is best accounted for by the current ecology of the animals rather than their phylogeny or motor prowess.

Sources of measurement error

However, the reliability of these relationships ultimately depends on the reliability of the estimates of the areas of the

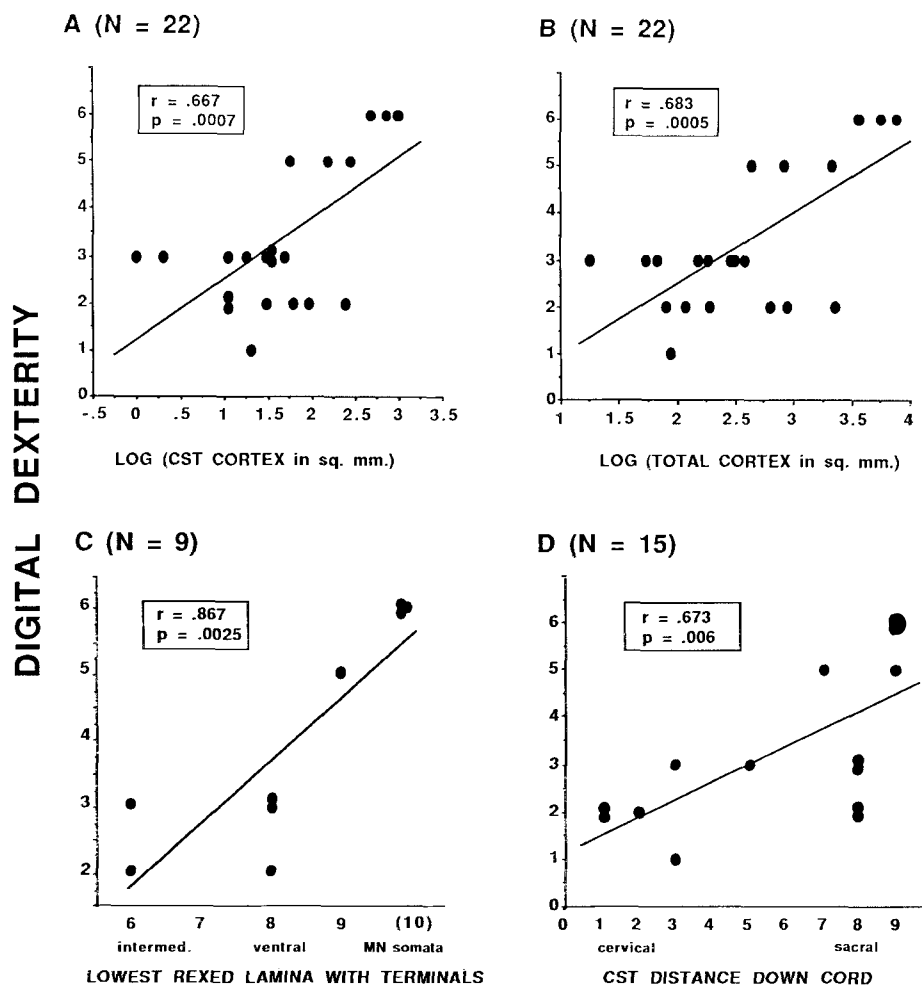


Fig. 7. Four best correlates of "Digital Dexterity" found among parameters describing morphology of CST. **Upper two graphs:** Based on present data describing the cortical origins of the CST. **Lower two**

graphs: Based on data of many others describing terminations of CST within spinal cord (see compilation in Heffner and Masterton, '75). Note highest correlation coefficient in C despite very small sample size.

cortical regions originating CST fibers and these estimates were subject to several sources of uncontrolled measurement error. Among these were errors in the labeling procedure itself; errors in the estimates of cortical areas from sets of histological sections; and finally, statistical error resulting from the non-random sampling of mammals.

The first of these sources of error, the error inherent in the cell-labeling technique, has been discussed previously (Nudo and Masterton, '89). Briefly, because of the histological screening procedures employed, it is more likely than not that the present materials err toward under-labeling rather than over-labeling the CST. That is, our procedures were designed more to eliminate "over-labeling" by the inadvertent labeling of cells which are *not* part of the CST (such as corticobulbar fibers) than "under-labeling" by the non-labeling of true CST fibers either by non-exposure to HRP or possibly, by an HRP-phobia of a subset of CST fibers.

For the present purpose, however, it should be noted that "under-labeling" of the CST due to the *random* non-labeling of some of its fibers would have an effect on the number and density of labeled cells in the cortex but it would have little effect on the extent of a labeled area. Hence, it would have little effect on the estimate of the area

of cortex originating CST fibers. Therefore, short of the possibility of not labeling entire and non-random subsets of corticospinal fibers, under-labeling error is probably the least important source of error in the present data.

The second type of measurement error inherent in the present technique is the one accompanying the estimate of an area based merely on subsets of histological sections through the labeled area. The error in this method of estimation is a direct function of intersection distance which is itself an indirect function of the number of sections per area measured. As the number of measured sections through a particular region increases (i.e., the intersection distance decreases) the estimate of the area converges rapidly on the "true value" of the area—i.e., the value obtained if every section was measured. In previous experiments, we have found that for any structure *not* sectioned parallel to a straight border (i.e., the first section and the last section are not relatively large values), the error falls to less than 5% with the measurement of eight equidistant sections through the structure and to less than 3% with the measurement of 11 equidistant sections (see Gundersen and Jensen, '87).

Because Region A and Region B, the two regions common to each of the species, were large enough to require 11 or

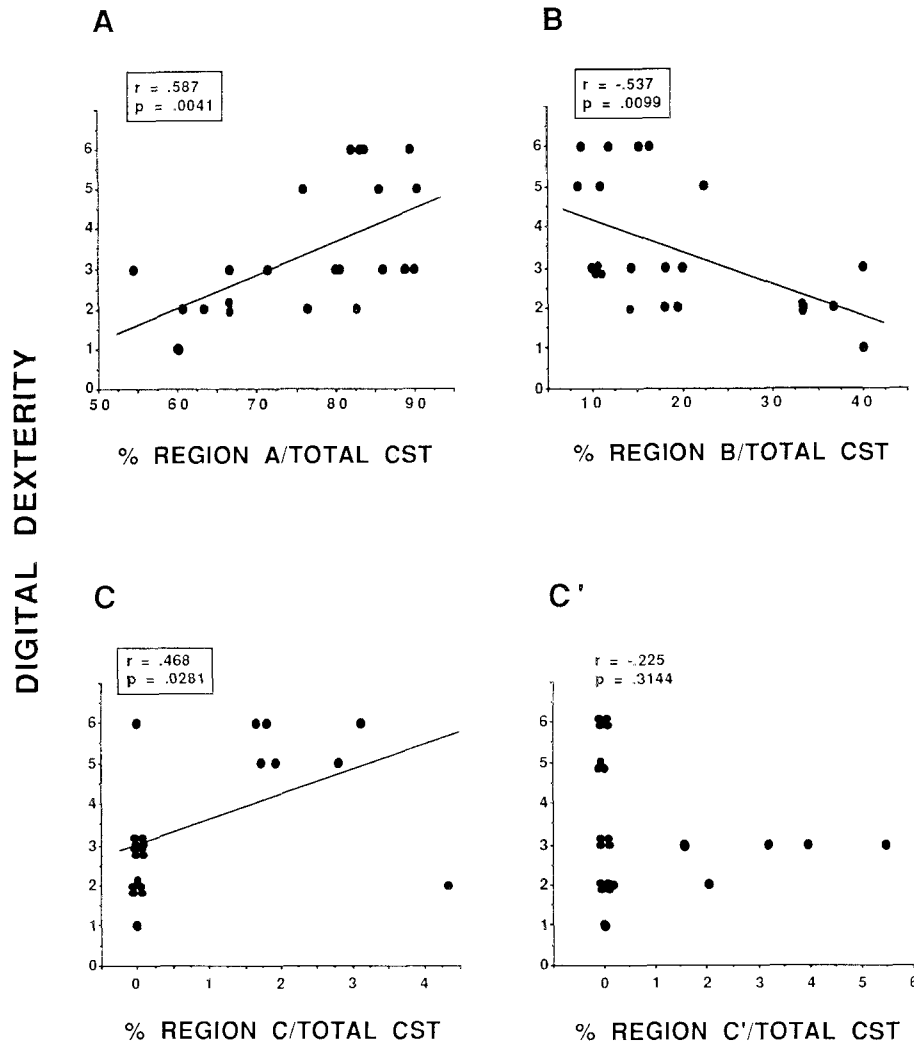


Fig. 8. Relationships of "Digital Dexterity" with relative size of the four cortical regions found in mammals (Nudo and Masterton, '89). Note statistically reliable *inverse* relationship (i.e., negative correlation and negative slope) with Region B. Only relative size of Region A

remains reliably related after variation of entire CST is mathematically removed. Therefore, despite its uniqueness to primates, Region C apparently added little or nothing to their dexterity.

more equidistant sections to span them, the measurement error of these regions is probably less than 3%. In the smaller brains however, and for Regions C and C' in most brains, one or another region was spanned by less than 11 sections. In these cases, the measurement error is higher than 3% though still less than 10%. Therefore, the error inherent in the raw data by measuring only sets of equidistant sections is probably less than 5% for Region A or B and less than 10% for Regions C and C'. The measurement error in the total area of cortex originating cortical spinal tract fibers (i.e., the sum of areas of the two or three regions) is also less than 10% and the error for the area of the entire neocortex is less than 3%.

However, because no correction was made for shrinkage due to fixation and histological processing, each of these measurements is biased toward values lower than *in vivo* values. This bias has been minimized, though not eliminated entirely, in the analyses here by using the raw measurements only in ratios or in correlation coefficients. Both of these derived values yield entirely unitless numbers which

are unaffected by relatively similar shrinkage of the cortical tissues within each animal.

Relationship of CST cortex to total cerebral neocortex and its historical constancy in mankind's lineage

Despite the large increase both in the absolute amount of cortex devoted to the CST and in the absolute amount of total cortex in mankind's ancestral lineage, the relative amount of CST cortex apparently remained almost constant over human ancestral lineage. The idea that this constancy in the relative amount of CST cortex is to be expected would seem to be contraindicated by obvious changes in each of the non-primate lineages represented in the sample. For example, the percentage of CST cortex more than doubles from the placental mammalian average along the raccoon's lineage. In contrast, the percentage of CST cortex decreases by 30% over the rodent segment of the tree squirrel lineage and along several other lineages as well. Therefore, the relative

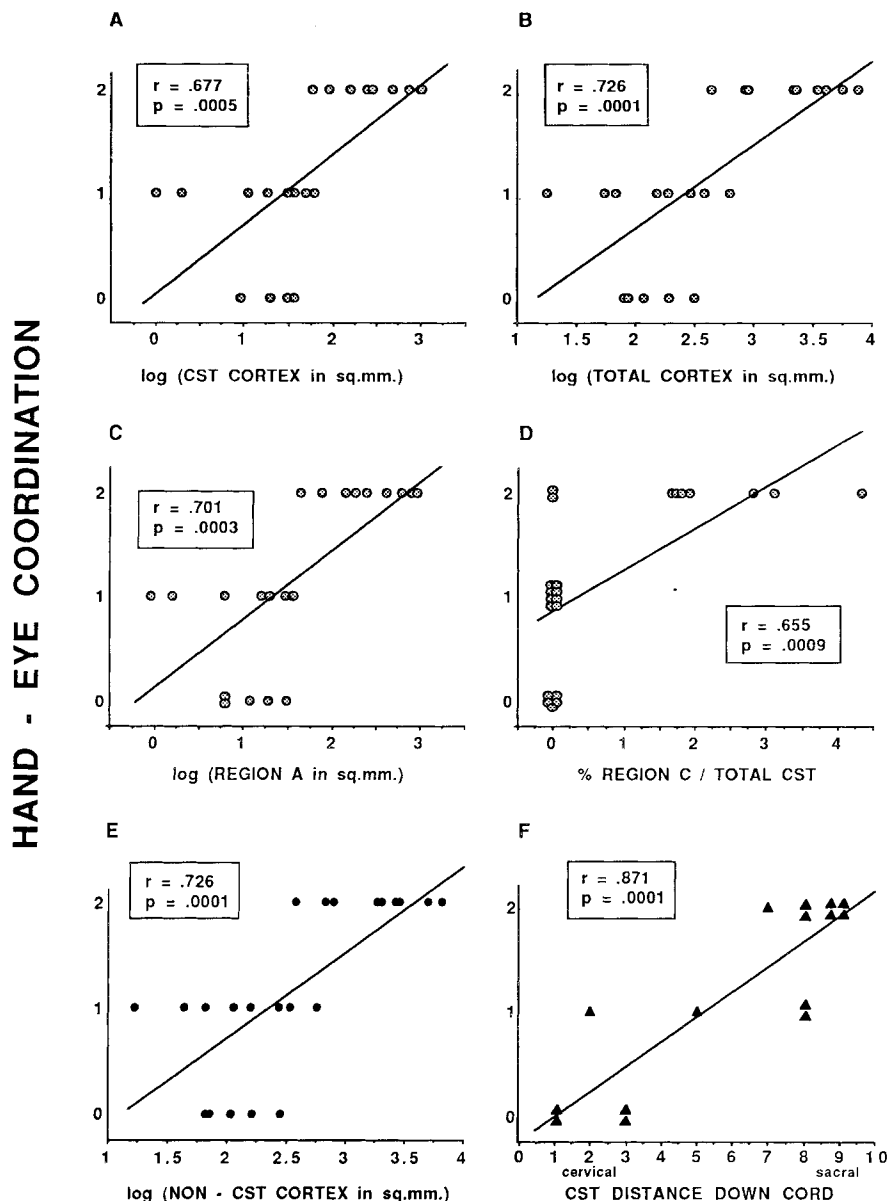


Fig. 9. A-E: Five best correlates of "Hand-Eye Coordination" found among present parameters describing cortical origins of CST. F: Correlate based on CST-termination data of others as compiled by Heffner and Masterton ('75). Note high reliability of relationships despite crude scaling of "Hand-Eye Coordination." The two relationships graphed in

E and F when taken together account for over 85% of the variance and entirely obviate the relationships shown in A-D. Therefore, the distance the CST extends down the spinal cord and "non-CST cortex" provide better clues for variation in "hand-eye coordination" than do the variations in CST cortex itself.

constancy of the percentage of CST cortex in mankind's lineage is not usual among mammalian lineages and consequently, it may contain a clue as to a possibly unique set of functions of the primate neocortex.

Since the amount of CST cortex remained as a relatively fixed proportion of total cortex over a very long time span in mankind's ancestral lineage, it follows that the amount of CST cortex and the total amount of neocortex are probably "yoked" in some anatomical or functional way.

If they are yoked, the question arises as to which tissue, the entire neocortex or only CST cortex, has the most direct selective pressure impinging upon it. For example, if selective pressure on the CST cortex is the driving force for its own expansion and yet the percentage of CST cortex

remains constant, then it follows that the expansion of the entire neocortex depends on, or is driven by, the expansion of CST cortex. This notion, in turn, suggests that *non-CST* cortex (that is, sensory and association cortex) may be little more than supporting tissue for an essentially "motor" cerebral hemisphere.

If, on the other hand, the selective pressure forcing changes in forebrain cortex is seen as primarily affecting the *total* neocortex (i.e., expanded sensory and association areas), the constancy of the CST percentage suggests that the absolute CST has been allowed to expand, or forced to expand, only up to a fixed proportion of the total. In this case, it is non-CST cortex which dominates forebrain function and the CST is allowed to expand perhaps only because

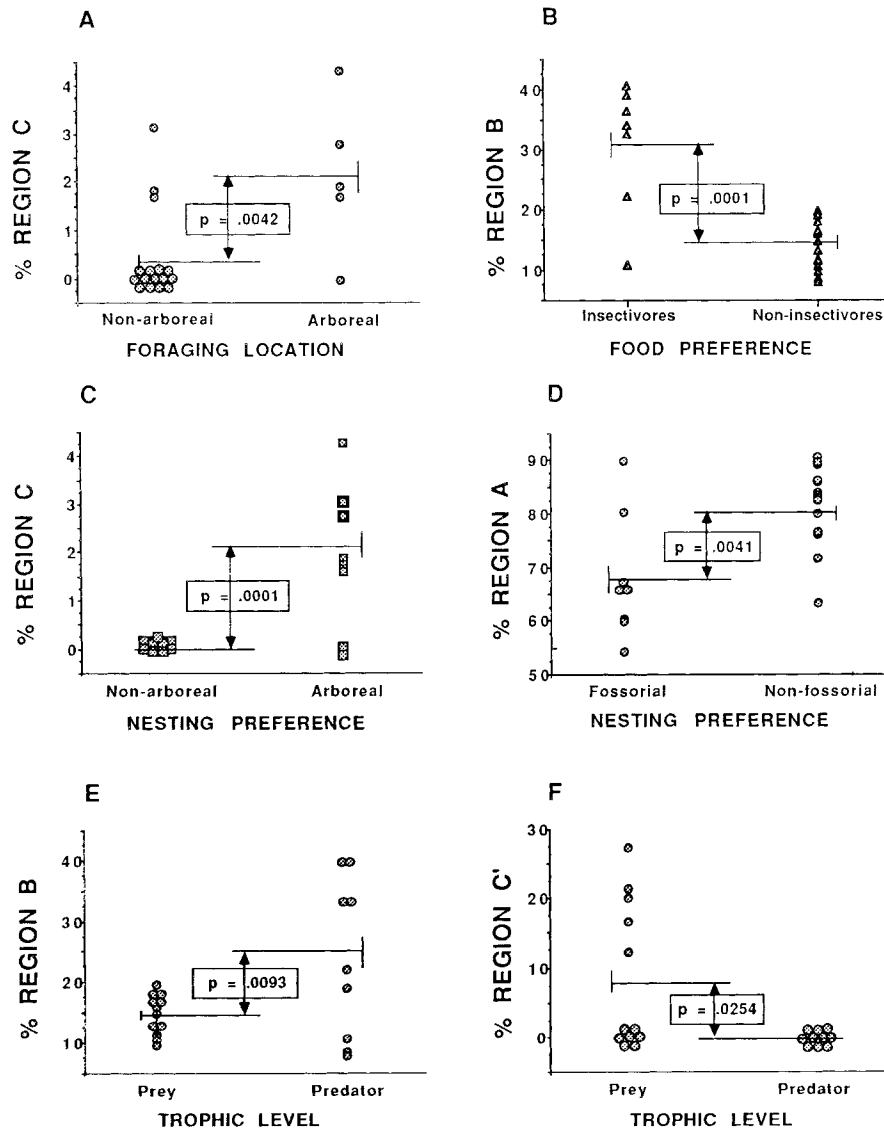


Fig. 10. Six statistically reliable relationships between ecological descriptors and the relative sizes of the specific cortical regions. In each graph horizontal lines are means and vertical lines are standard errors of the means for each of the specified subgroups of the animals. Boxed "p"

is probability of the difference between the two means occurring by chance alone. Note that relative size of Region C (the one region possibly unique to primates) is more closely related to arboreal habitat (graphs A and C) than to digital dexterity (Fig. 8C).

it is the "output device" of an essentially "sensory" or "associative" cerebral hemisphere.

In either event, the relative constancy of CST percentage in the primate lineage along the long time period spanned by the common ancestry of the animals in the present sample while the absolute amounts of both CST cortex and total cortex were expanding exponentially suggests an unusually close and possibly cause and effect relationship between the two tissues. Although the present data do not point clearly to either one as the more primary (or causal or limiting) variable, the closeness of the relationship and its persistence throughout a *very* long period of time suggest that one of the two may indeed be causal in the usual sense of that term.

Distribution of CST origins among cortical regions

Although the results show a relatively constant percentage of cortex devoted to the CST along the anthropoid

lineage (Fig. 5), the regions within CST cortex are not constant. The data suggested that there was probably an increase both in absolute and relative amounts of cortex within Region C, whereas Regions A and B increased in absolute size but probably remained constant or even decreased in relative size over the same time period (Fig. 6). Because Region C seems to be unique to the primates, it warrants special comment.

It is tempting to presume that the appearance of Region C in prosimians and its subsequent expansion in absolute size among the anthropoids might account for the increases in primate motor prowess in general and digital dexterity in particular. However, this idea is unsupported by the present data.

To begin with, Region C probably expanded in absolute size over the primate lineage, but its size relative to total CST cortex may have shown no progressive expansion at all. Figure 6 shows that Region C may have appeared with the

origin of prosimian primates and then remained constant in relative size thereafter. If this was the case, it is difficult to argue that Region C, by itself, is the substrate for any progressive acquisition of motor skills shown in the later stages of the primate lineage.

Indeed, the present data suggests that among the three regions present in primates, Region C may be the *least* related to gains in digital dexterity. Figure 8C shows that among primates alone (i.e., see only the seven points with a non-zero abscissa in Fig. 8C), the relative size of Region C is *inversely* related to digital dexterity. Therefore, the dexterity of the Old World monkeys may be related to the gains in the absolute amount of cortex devoted to the CST but it is not obviously related either to Region C or to the distribution of the CST fiber origins among the three regions of CST cortex. In sum, the relationships of the individual regions to motor prowess, whether digital dexterity or hand-eye coordination, are relatively weak and of arguable reliability.

Instead, the distribution of CST fiber origins among the regions seems to be most closely related to current ecological adaptations. For example, Figure 10 shows that both an arboreal nesting preference and an arboreal foraging location are the best predictors of the relative size of Region C. Since both of these modes of adaptation are characteristics of primates and since primates are the only members of the sample with a Region C, the data indicates that it is the arboreal life-style of primates that may have provided the selective pressure for the origin and expansion of Region C. At the least, this correlate of Region C is much stronger than its correlate with "digital dexterity" (Fig. 8) and also stronger than its correlate with "hand-eye coordination" (Fig. 9). In seeking a functional correlate of Region C, therefore, it would seem at present that the best clue is to be found in some motor trait necessitated by an arboreal life and more closely related to hand-eye coordination than to digital dexterity.

Similarly, the relative sizes of Regions A and B are more closely related to trophic level, nesting preference, or food preference than they are to dexterity. Although we know of no single dimension that combines these factors into a simple explanatory idea, the relatively high reliabilities

shown in Figure 10 suggest that they contain a clue to function that probably should not be overlooked. Although the bases of these relationships are not now obvious to us, they might well become clearer as data from more widely varied mammals is added to the present pool.

ACKNOWLEDGMENTS

The authors thank Dr. O.B. Mock, Kirksville College of Osteopathic Medicine, for providing us with the least shrews (*Cryptotis parva*), and Dean Sutherland and Steve Suknaic for helping to plot the HRP-labeled cells in dozens of specimens.

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